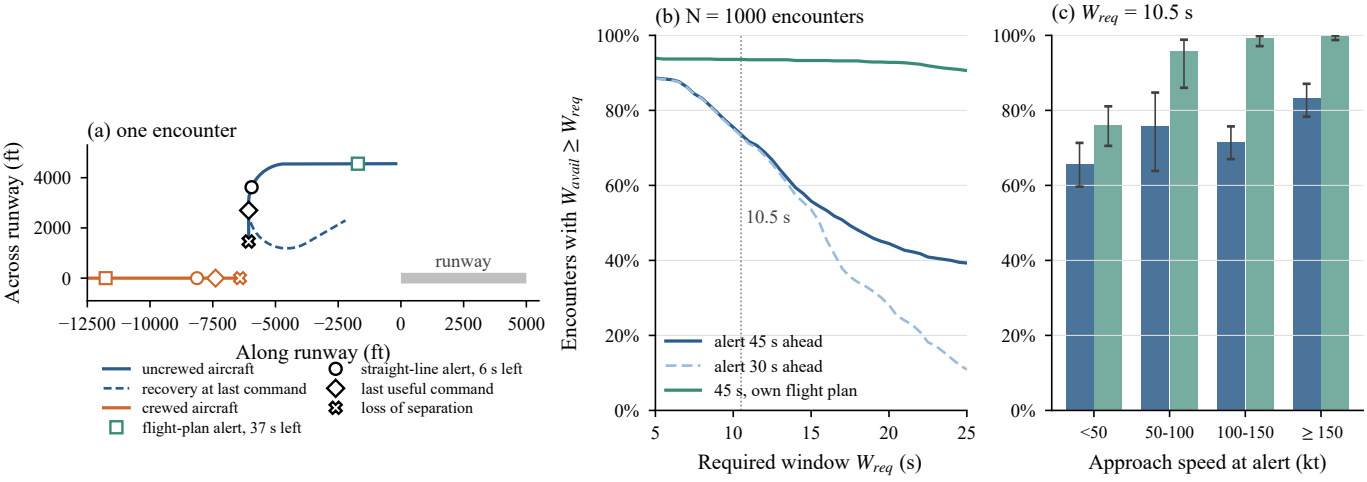
(a) one encounter
Across runway (ft)
4000
2000
0
−12500 −10000 −7500 −5000 −2500 0 2500 5000
Along runway (ft)
runway
uncrewed aircraft
recovery at last command
crewed aircraft
flight-plan alert, 37 s left
straight-line alert, 6 s left
last useful command
loss of separation
(b) N = 1000 encounters
Encounters with $W_{avail} \geq W_{req}$
100%
80%
60%
40%
20%
0%
10.5 s
alert 45 s ahead
alert 30 s ahead
45 s, own flight plan
5 10 15 20 25
Required window W_{req} (s)
(c) W_{req} = 10.5 s
100%
80%
60%
40%
20%
0%
<50
50-100
100-150
≥ 150
Approach speed at alert (kt)